

EVO⁶⁴ Long Live the EVOLUTION

Inspired by Assy 250407
EPROM Based Multi-ROMs
ESD Protection on all Ports
Dual-Crystals for PAL & NTSC
Integrated RF Bypass
Supports all SID & VIC-II models

1.0



2.0

Integrated ClearVideo64c circuit
Jitter free VCO Clock circuit
Reduced noise in all power & data lines
Began transition to SMD components

3.0

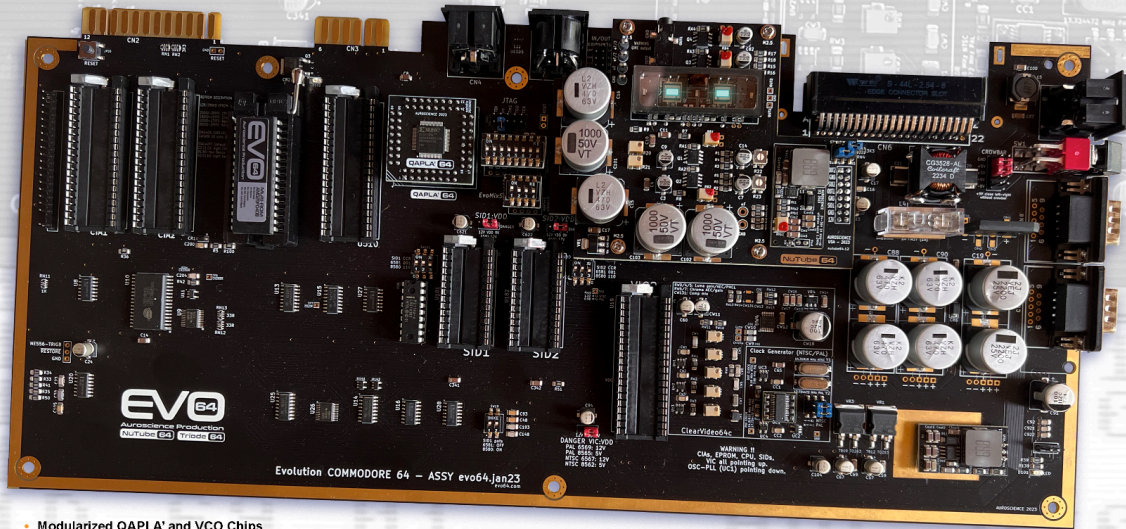
Dual-Stereo SIDs & audio amps
Unified single-EEPROM multi-ROM
CPLD-based PLA
Further transition to SMD components
Additional signal noise reduction

3.4

Transition to SRAM for main memory
Extensive audio signal smoothing
First generation audio preamp modules
Further transition to SMD components

3.5

Transition to SMD logic completed
Premanufacturing for quick build
Second gen audio preamp modules



- Modularized QAPLA' and VCO Chips
- Chips repositioned to better accommodate ZIF sockets
- Micro-DIP switches have replaced Jumpers for fixed functions
- ESD Chips relocated to accommodate more keyboard mounting options
- Optional 3rd generation audio preamp modules available
- Joystick port standoffs eliminated to ensure the best fit into most case styles
- Optional Colcraft AC Filter Choke and Ultra Low Noise Switching voltage regulator

READY.



EVO64 was developed out of the interest to combine many of the best features that have become available for the Commodore 64, into a sleek and fully integrated system. EVO is made using modern manufacturing processes, but care has been taken to ensure that it remains completely true to the original system's heritage and architecture.

For starters, EVO delivers the best retro video experience available, through its integrated [ClearVideo64c](#) circuit, which provides vibrant colors and none of those distracting "jail bars" (inspired by work done by Jay Desmarais). Irrespective of whether your display supports a Composite or S-Video signal, both output formats are fully tunable, using a set of high precision potentiometers for micro adjustments. You'll get the clearest images and the sharpest text that you can imagine.

EVO was designed with compatibility in mind. You're free to use *any* VIC-II chip version that you prefer, from the old ceramic 6567 (NTSC) or 6569 (PAL) chips, which required 12v power, to the later generation 8562 (NTSC) or 8565 (PAL) versions, which switched over to 5v. It's up to you. In fact, you can easily switch between them, with adjustable jumpers to set voltages and to switch from PAL to NTSC video modes.

Similarly, you can choose *any* SID chip variant(s) that you prefer, from the earliest generation 6581, with its gnarly oscillators and filter glitches (operating at 12v), to the clean and pristine 8580 (operating at 9v). And as you would expect, EVO is also fully compatible with the most popular SID clones, like the ARMSID, the FPGA SID, SwinSID, SwinSID Ultimate, SIDKick, BackSID, and others. Best of all, you can mix and match any combination of SIDs that you like, to get the best version of that retro sound you've been hearing in your head.

To ensure the best [Dual-SID compatibility](#), EVO incorporates logic that has been inspired by the MixSID (originally developed by Henning Liebenau). This enables a second SID to operate in either a dual-mono or full stereo mode. The primary SID will always listen on address \$D400, which ensures 100% backwards compatibility, however the second SID can be configured to either mirror the primary, or receive its own notes and instructions from *both* \$D420 and \$D500. This enables support for hundreds of the most amazing sounding dual-SID tracks from sources like the [High Voltage SID Collection \(HSVC\)](#), as well as great dual-SID capable games, like the new Mario Brothers clone. You can also make your own stereo music with composer software like the CYNTHCART. A simple jumper adjustment will allow the second SID to listen on additional addresses, like \$DE00 and \$DF00 as well, which provides further compatibility, for tools like the MSSIAH music composition cartridge, and other music players.

And for the "tried and true" SID music fans out there, EVO has some *special* surprises in the form of two custom developed [Audio Preamp Modules](#), which mount directly onto the main board. This ensures that the cleanest and lowest-noise signals are sent into the preamps, enabling them to provide unique warmth, color and sound character, unlike *anything* you've ever heard from a C64.

[EVO ROMs](#) are fully reprogrammable and instantly swappable too. In fact, a single EPROM (the W27C010) can provide support for up to 8x different Kernel ROM images, as well as 4x different versions of Basic and 8x character sets. Bring along your own JiffyDOS or JaffyDOS (must be licensed to use). Experiment with blazing-fast load times, using DolphinDOS or SpeedDOS, etc. Switching between ROMs is as easy as moving a jumper cap or tapping a *'hot-key combination'* using the [EVO Multi-Switch Module](#). With larger EPROMs (e.g., a 29F040 or 27C080, etc.) EVO can store Cartridge images as well. This provides convenient access to the software you use most, like testing and diagnostics, or even music synthesizers, trackers, SID Players, and of course games!

Every EVO system comes with a proprietary PLA chip that we call [the \(QAPLA'\)](#). This incorporates a series of logic algorithms that have been extensively tested against the most highly-compatible solutions currently available, such as the neatPLA, the PLAnkton, and others. There's no need to worry about sourcing one of those increasingly rare and known to be volatile PLA chips, like the MOS 906114-01. There's also no need to pay big bucks for a giant SuperPLA chip like the 252535-01. You can be confident that your games, demos, cartridges and add-ons will work flawlessly.

EVO also has electrical static discharge [\(ESD\) protection](#) for the joystick and I/O ports. This prevents static electricity and reverse voltages from damaging your system and those precious MOS/CSG chips.

Most importantly, EVO is an *excellent* hands-on DIY project that's simply [Fun to Build](#). Most of the components from the original C64 have been modernized to use newer, low profile, surface mount equivalents (SMD/SMT), which come pre-soldered onto the board. This makes it quick-and-easy to build an EVO and even to desolder and replace parts by hand, if that's ever needed. There are multiple test points, strategically placed all around the PCB for your multimeter and oscilloscope, as well as other conveniences, like an integrated tactile reset button that you can press while taking measurements (saves you from having to power-cycle, etc.). There's even a set of SMD LEDs that to indicate the system's power status, and dataset access, while the board is outside of a case. All of this makes EVO easy to work with and convenient to perform testing and diagnostics with, and best of all it invites your creativity to design your own mods & upgrades.

Despite all the features and flourishes, EVO is still a fully authentic C64 reimplementation. It's backwards-compatible with all of your original MOS chips and popular clones, although these are the *only* original parts that you'll actually need to source. *Everything else* is already integrated.

Once you've experienced the EVO difference, we think you'll agree that *"Retro has EVOLved"*...

Jan Beta



Jan Beta



Retro Recipes



Retrobits



[CLICK HERE IF YOU WOULD LIKE TO ORDER AN EVO:](#)

◀ Retrospective

Acknowledgements:

- Technical engineering and PCB layout by [Dr. Stefano \(aka Auro\)](#). Includes the integration and synthesizing of multiple concepts, into a completely new but still quite familiar system architecture
- EVO is loosely based on a combination of Commodore 64 designs, ranging from the original assembly 250407 (log board) through to the later generation assembly 250469 (shortboard), copyrights belonging to Commodore Business Machines
- Inspiration for early EVO64 prototype designs were based on work done by [Hans Petter Dalsklev](#) on his reverse engineering of the Assembly 250407
- EVO's video circuit is inspired by work done by Jay Desmarais ([Jaystonian](#)) on the ClearVideo64c
- EVO's Multi-SID and multi-ROM circuits were inspired by work originally done by [Henning Liebenau](#) on the MixSID and RePROM respectively. Although, EVO's implementations have been heavily modified from the originals
- Voltage Controlled Oscillator (VCO) / MOS 8701 Clock replacement circuit was inspired by designs created Jeff Birt ([HeyBirt](#))
- EVO64 logo, case badges and other branding have been developed by the [Retro Badgeman](#)
- Social media, marketing, tech documentation, website support, "innovative ideation", as well as beta builds & tech testing by [D. Westbury](#)
- Additional beta builds & testing by [M. Bernardo \(aka SX64Man\)](#) and H. Eckert.